

Rutik Dattatray Avhad

Pune, Maharashtra

rutikavhad2211@gmail.com — +91 8767689952 — [LinkedIn](#) — [GitHub](#) — [Portfolio](#)

Summary

M.Sc. Computer Science student with a robust foundation in full-stack web development, database design, and cybersecurity. Completed coursework and hands-on projects in backend development, API design, and web application security. Eager to apply analytical and problem-solving skills to build secure, scalable software solutions and contribute to performance-driven development environments.

Education

M.Sc Computer Science 2024 – 2026
PUCSD Savitribai Phule Pune University. Pune

B.Sc Computer Science 2021 – 2024
R. B. Narayanrao Borawake College. Shirampur

Skills

Programming Languages: Python, Java, C, SQL.

Databases: PostgreSQL, MongoDB

Backend & Frontend: Django, Flask, SQLAlchemy, REST API, HTML, CSS, Bootstrap.

Core Concepts: Data Structures & Algorithms, OOP, SDLC, Debugging.

AI / Machine Learning: Face Recognition, Computer Vision, InsightFace, OpenCV

Systems: Linux, Windows.

Technical Skills & Tools: Networking, Operating Systems, Git, GitHub, Docker, VS Code, Pandas, Selenium, BeautifulSoup, JWT, Postman, GitHub Actions.

Projects

IntelliWAF – Intelligent Web Intrusion Detection & Prevention System. [GitHub](#) Sep 2025 – Dec 2025

- Architected a dual-engine platform featuring a high-performance Python/Mitmproxy request-interception pipeline alongside an independent Flask full-stack administration dashboard.
- Developed request interception and payload normalization pipelines to analyze incoming HTTP/HTTPS headers, cookies, and POST request bodies for enhanced security evaluation.
- Achieved 97% attack detection accuracy during testing against DVWA in a Metasploitable lab environment.
- Validated detection accuracy against DVWA using over 70+ attack payloads.

Technologies: Python, Mitmproxy, Flask, HTML, CSS, REST API, Kali Linux, Metasploitable DVWA Lab.

FaceHR: AI Attendance & Payroll System. [GitHub](#) Apr 2026 – Jun 2026

- Developed a full-stack HRMS using Flask, PostgreSQL, SQLAlchemy, and Flask-Login for employee, attendance, leave, and payroll management
- Implemented AI-based face recognition attendance using InsightFace and OpenCV for automated employee check-in/check-out.
- Built payroll automation with salary calculation based on attendance hours, paid leave policies, payment tracking, and QR-based salary payment workflow.
- Integrated secure authentication, role-based access control, face registration, attendance kiosk system, and real-time reporting features.

Technologies: Python, Flask, PostgreSQL, SQLAlchemy, Flask-Login, OpenCV, InsightFace, HTML, CSS, JavaScript, Bootstrap, Pandas, OpenPyXL, QR Code API.

Authentication & Access Control Vulnerability Scanner. [GitHub](#) Jan 2026 – May 2026

- Built automated security scanner detecting broken authentication and authorization flaws.
- Implemented endpoint discovery by crawling target URLs and extracting all forms/API endpoints.
- Added vulnerability checks for SQL/NoSQL injection, rate limiting, CSRF tokens, and JWT attacks.
- Generated detailed JSON reports with severity levels (High/Medium/Low) and remediation suggestions.

Technologies: Python, Web Security Testing, Docker, Selenium, BeautifulSoup, Automation .

Pune Metro Live Train Tracker. [GitHub](#) Jan 2026 – May 2026

- Built train movement simulation covering Pune Metro routes with real-time position tracking.
- Designed normalized PostgreSQL schema with multi tables including stations, routes, schedules, and train timings, etc.
- Implemented conflict prevention algorithm managing 3-4 simultaneous trains across 15km stretch.
- Developed backend routing and frontend interfaces using Flask, HTML, and CSS.

Technologies: Python, Flask, PostgreSQL, HTML, CSS.